

KUSTENDIL, BulgariaWMO: **156010**Lat: **42.267N**Lon: **22.767E**Elev: **1729**StdP: **13.80**Time Zone: **2.00 (EUE)**Period: **94-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	14.8	4.9	7.5	11.4	10.1	9.8	17.5	23.0	35.2	19.5	34.2	0.9	N/A	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	92.5	67.8	89.1	67.2	86.1	66.3	70.3	87.0	68.9	85.0	67.5	82.9	3.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.5	96.9	78.6	63.0	91.9	76.1	61.9	88.2	74.5	35.4	87.3	34.0	85.3	32.9	83.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	15.1	11.5	4.5	97.5	6.2	3.9	0.0	100.3	-3.6	102.6	-7.0	104.7	-11.5	107.5
			3.9	73.6	5.8	1.8	-0.3	74.9	-3.6	76.0	-6.9	77.0	-11.0	78.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	32.0	35.7	43.9	52.0	60.6	67.0	71.6	71.6	63.2	53.8	42.4	33.5	(6)	(7)
	DBStd	15.65	8.23	8.30	7.52	6.47	5.76	5.70	4.96	4.89	5.80	7.09	7.69	9.18	(7)	(8)
	HDD50	2037	557	403	213	51	2	0	0	0	1	45	249	516	(8)	(9)
	HDD65	5208	1022	821	653	392	157	44	7	6	100	351	678	977	(9)	(10)
	CDD50	2901	1	1	25	110	331	509	669	670	398	162	21	4	(10)	(11)
	CDD65	596	0	0	0	1	21	102	211	211	47	3	0	0	(11)	(12)
	CDH74	6869	0	0	1	33	345	1194	2328	2310	591	66	1	0	(12)	(13)
	CDH80	2594	0	0	0	5	62	392	956	991	181	7	0	0	(13)	(14)
(14) Wind	WSAvg	2.4	2.0	2.8	3.7	3.2	2.5	2.3	2.6	2.2	2.0	1.6	1.9	2.0	(14)	(15)
(15) Precipitation	PrecAvg	23.1	1.4	1.6	1.6	2.0	2.4	2.5	1.9	1.8	1.8	2.1	1.8	2.3	(15)	(16)
	PrecMax	29.4	3.5	4.0	3.3	4.6	4.7	4.8	4.1	4.3	4.7	4.8	3.8	5.7	(16)	(17)
	PrecMin	13.0	0.2	0.1	0.2	0.6	1.4	0.4	0.3	0.3	0.6	0.2	0.0	0.6	(17)	(18)
	PrecStd	4.7	0.9	1.0	0.8	0.9	0.9	1.2	1.0	1.2	1.0	1.3	0.9	1.2	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	62.7	70.7	78.7	85.4	91.8	97.3	96.7	89.8	80.1	68.2	60.8	(19)	(20)
		MCWB	47.4	49.7	54.1	59.3	64.6	68.9	67.9	68.5	65.3	62.6	56.7	53.2	(20)	(21)
	2%	DB	50.8	57.1	65.8	72.2	81.2	88.0	92.4	92.9	85.2	75.2	62.9	55.4	(21)	(22)
		MCWB	44.4	46.8	51.1	55.2	62.8	68.2	67.8	67.9	64.0	60.0	54.0	48.7	(22)	(23)
	5%	DB	46.6	52.5	61.9	68.3	77.9	84.5	88.9	89.3	80.9	70.7	58.1	50.6	(23)	(24)
		MCWB	41.6	44.7	49.3	53.7	62.0	67.1	67.4	66.7	62.5	58.0	51.4	46.6	(24)	(25)
	10%	DB	43.1	48.3	57.7	64.9	74.4	81.0	85.4	85.8	76.4	66.7	54.5	46.3	(25)	(26)
		MCWB	39.4	42.3	47.3	52.3	60.6	65.6	66.6	65.9	60.7	56.3	48.9	43.1	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	50.2	55.2	59.8	66.9	71.7	72.4	72.5	68.1	63.8	57.5	54.3	(27)	(28)
		MCDB	55.6	61.2	69.9	76.0	81.1	86.8	90.2	90.3	83.6	78.0	66.6	60.1	(28)	(29)
	2%	WB	45.0	47.3	52.1	56.7	64.2	69.7	70.2	70.1	65.2	61.2	54.7	49.4	(29)	(30)
		MCDB	50.0	55.7	64.1	69.2	77.6	85.2	86.8	87.8	81.2	72.0	61.1	54.2	(30)	(31)
	5%	WB	42.2	45.0	49.8	54.8	62.6	68.0	68.6	68.3	63.5	59.2	52.3	46.7	(31)	(32)
		MCDB	45.9	52.0	60.4	66.0	76.0	82.5	84.6	85.4	77.6	68.7	57.4	51.0	(32)	(33)
	10%	WB	39.6	42.6	47.7	53.1	60.9	66.2	67.3	66.8	62.0	57.2	49.5	43.2	(33)	(34)
		MCDB	42.9	48.0	56.6	63.3	73.4	79.6	82.5	82.7	73.8	65.3	54.1	46.1	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	17.6	19.9	20.3	22.0	23.3	24.9	25.4	23.3	21.5	16.9	12.1	(35)	(36)
		MCDBR	18.8	26.7	28.6	28.3	29.3	30.4	31.7	32.3	31.6	28.4	22.1	16.2	(36)	(37)
	5% WB	MCWBR	12.7	17.1	16.0	14.3	13.6	13.0	11.6	11.8	13.2	14.8	14.1	11.2	(37)	(38)
		MCWBR	16.1	24.7	26.6	24.7	26.7	27.7	27.5	28.7	27.1	24.7	19.8	15.3	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.313	0.341	0.384	0.441	0.434	0.424	0.424	0.437	0.412	0.374	0.341	0.315	(39)	(40)
		taud	2.387	2.318	2.238	2.091	2.183	2.249	2.250	2.213	2.264	2.359	2.413	2.410	(40)	(41)
	Ebn at Noon		262	271	271	262	268	270	269	261	259	256	248	250	(41)	(42)
		Edh at Noon	25	32	39	49	46	43	43	43	38	30	25	23	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		521	795	1119	1420	1713	1964	2039	1827	1346	948	611	449	(43)	(44)
	RadStd		79	103	128	169	134	165	143	147	111	115	61	71	(44)	(45)